



Wired LANs: Ethernet

Dr. Alekha Kumar Mishra

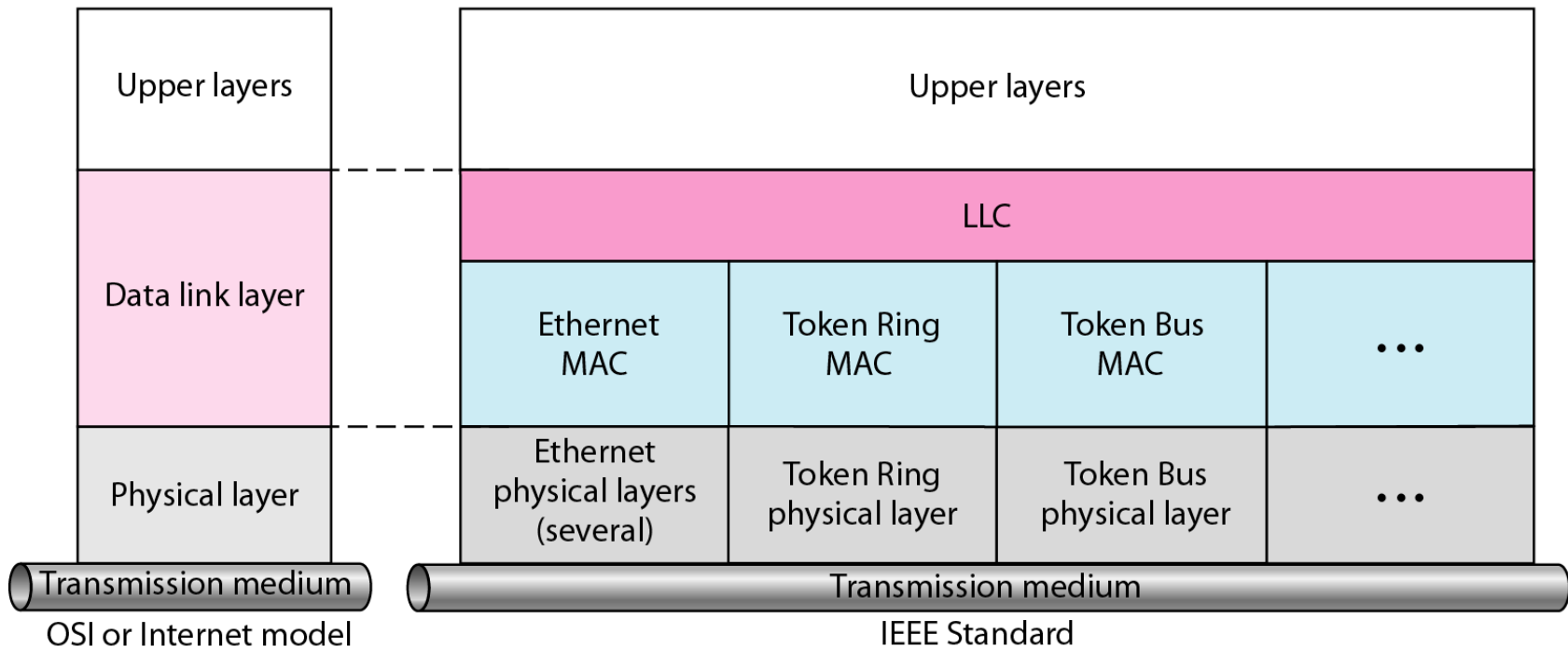


The Ethernet Standard

- Started in 1985 as **Project 802** by the Computer Society of the IEEE
- Objective to set standards to enable intercommunication among equipment from a variety of manufacturers
- It is a way of specifying functions of the physical layer and the data link layer of major LAN protocols.
- It was adopted by ANSI
- In 1987, the ISO also approved it as **ISO 8802**.

IEEE standard for LANs

LLC: Logical link control
MAC: Media access control



- The IEEE has subdivided the data link layer into two sublayers:
 - logical link control (LLC)
 - media access control (MAC)



Logical Link Control (LLC)

- In IEEE Project 802, **flow control**, **error control**, and part of the **framing** duties are collected into one sublayer called the logical link control
- Framing is handled in both the LLC sublayer and the MAC sublayer
- Framing LLC defines a protocol data unit (PDU) that is somewhat similar to that of HDLC
- The purpose of the LLC is to provide flow and error control for the upper-layer protocols that actually demand these services



Media Access Control (MAC)

- IEEE Project 802 has created a MAC sublayer that defines the specific access method for each LAN
- It defines **CSMA/CD** as the media access method for Ethernet LANs and the **token-passing** method for Token Ring and Token Bus LANs
- In contrast to the LLC sublayer, the MAC sublayer contains a number of distinct modules; each defines the access method and the framing format specific to the corresponding LAN protocol



STANDARD ETHERNET

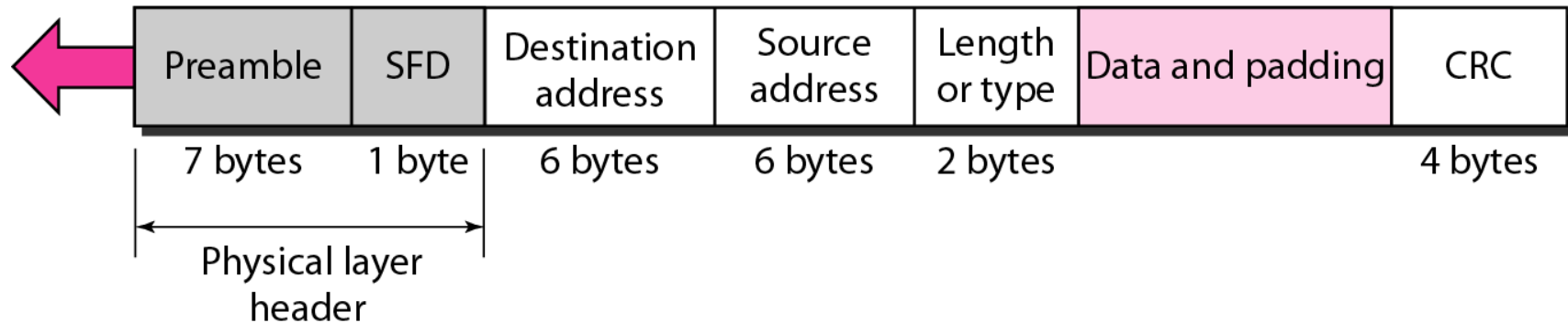
- The original Ethernet was created in 1976 at Xerox's Palo Alto Research Center (PARC)
- Four generations:
 - Standard Ethernet (10 Mbps)
 - Fast Ethernet (100 Mbps)
 - Gigabit Ethernet (1 Gbps)
 - Ten-Gigabit Ethernet (10 Gbps)



802.3 MAC frame

Preamble: 56 bits of alternating 1s and 0s.

SFD: Start frame delimiter, flag (10101011)



- The minimum length of the frame is 64 bytes
- The maximum length of the frame is 1518 bytes
- Each station on an Ethernet network has its own network interface card (NIC). The NIC fits inside the station and provides the station with a 6-byte physical address

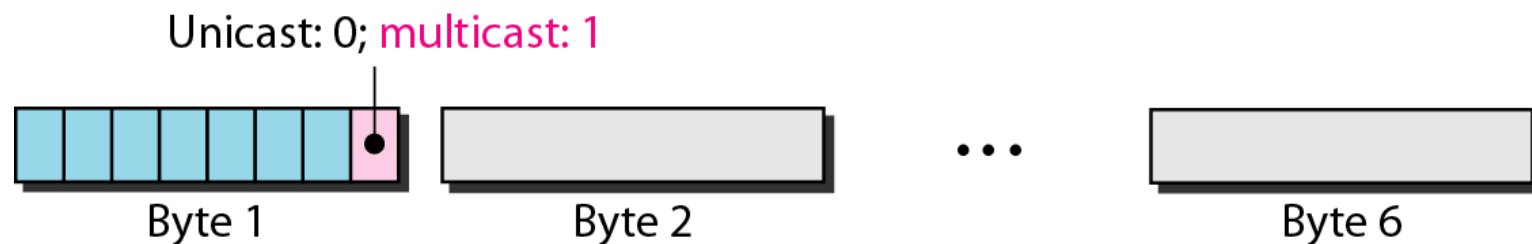
06 : 01 : 02 : 01 : 2C : 4B

6 bytes = 12 hex digits = 48 bits



Unicast, Multicast, and Broadcast Addresses

- If the least significant bit of the first byte in a destination address is 0, the address is unicast; otherwise, it is multicast
- The broadcast destination address is a special case of the multicast address in which all bits are 1s.



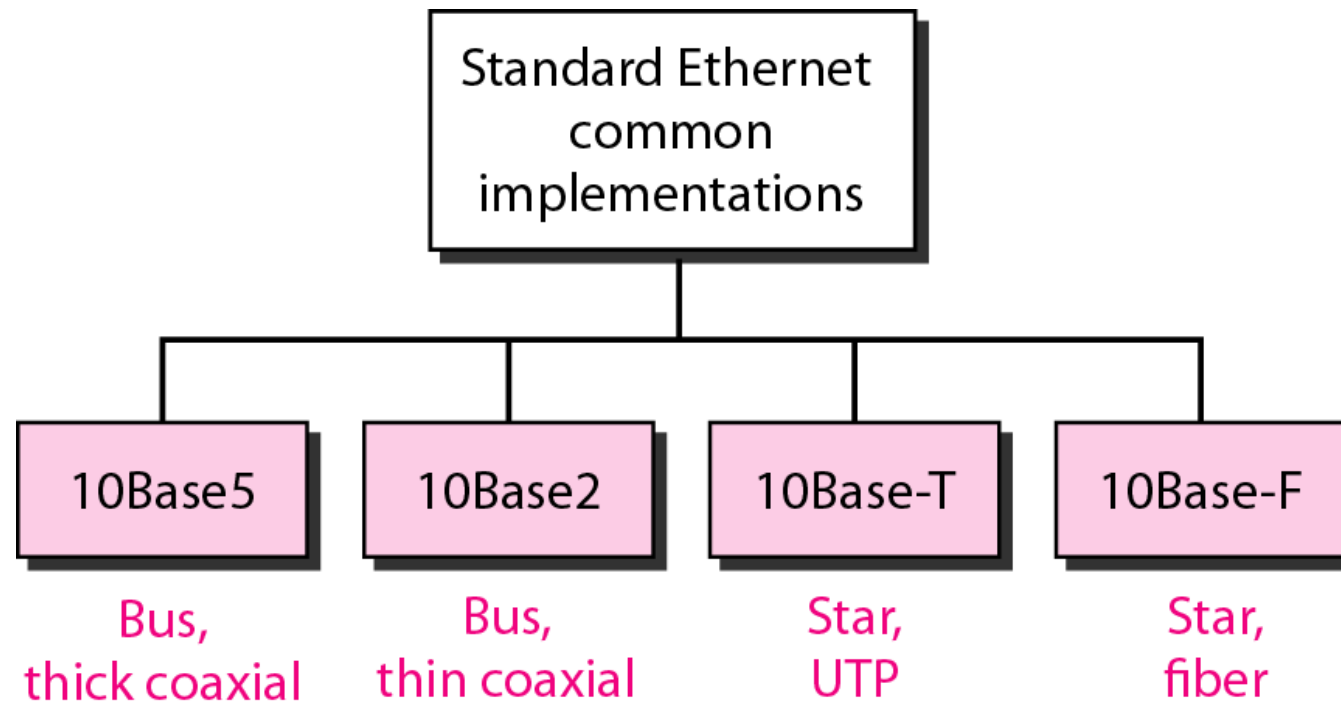


Access Method: CSMA/CD

- Standard Ethernet uses **1-persistent CSMA/CD**
- Slot time = round-trip time + time required to send the jam sequence
- The slot time in Ethernet is the time required for a station to send 512 bits
 - For traditional 10-Mbps Ethernet it is 51.2 microseconds

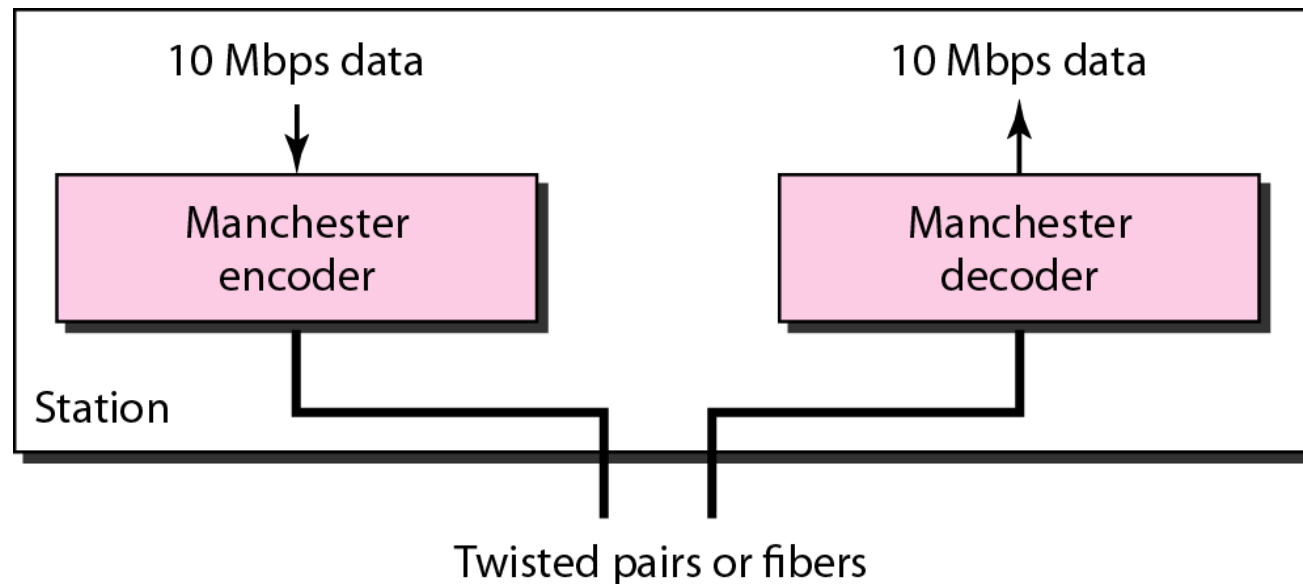


Categories of Standard Ethernet



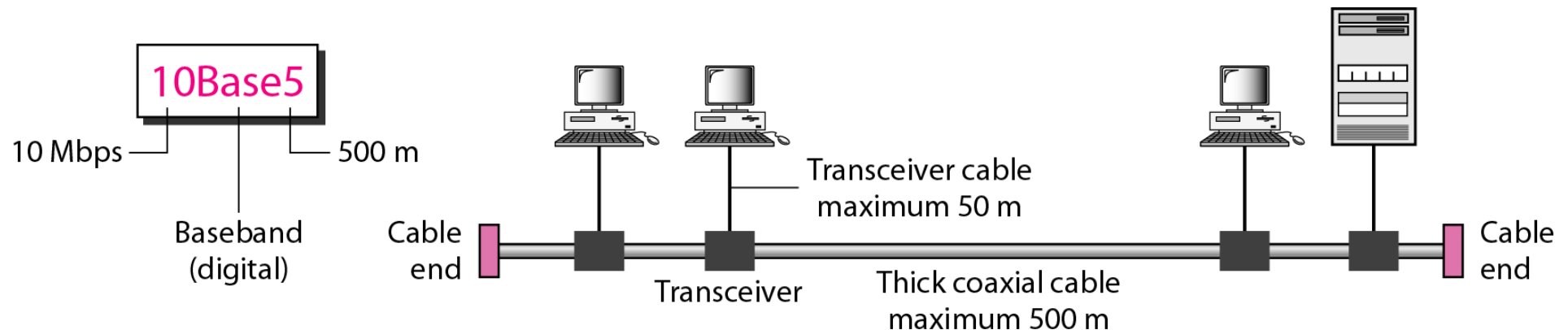
Encoding and Decoding

- All standard implementations use digital signaling (baseband) at 10 Mbps
- At the sender, data are converted to a digital signal using the **Manchester scheme**



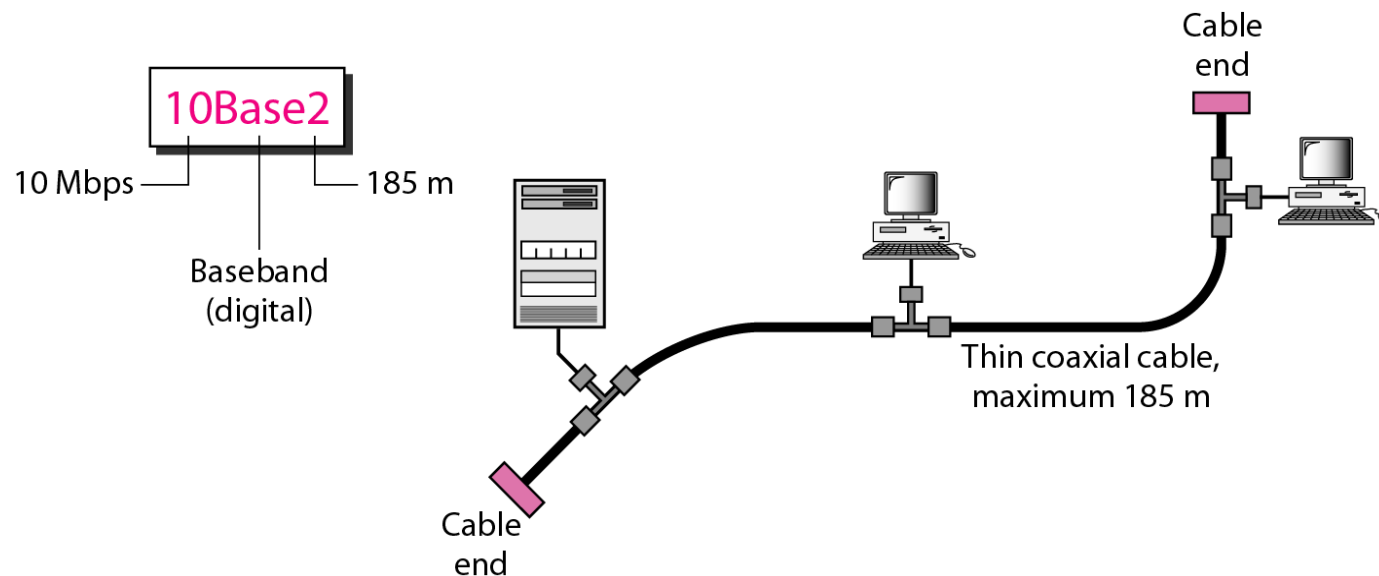
10Base5: Thick Ethernet

- Use a **bus topology** with an external transceiver (transmitter/receiver) connected via a tap to a thick coaxial cable
- The transceiver is responsible for transmitting, receiving, and detecting collisions.
- The maximum length of the coaxial cable must not exceed **500 m**



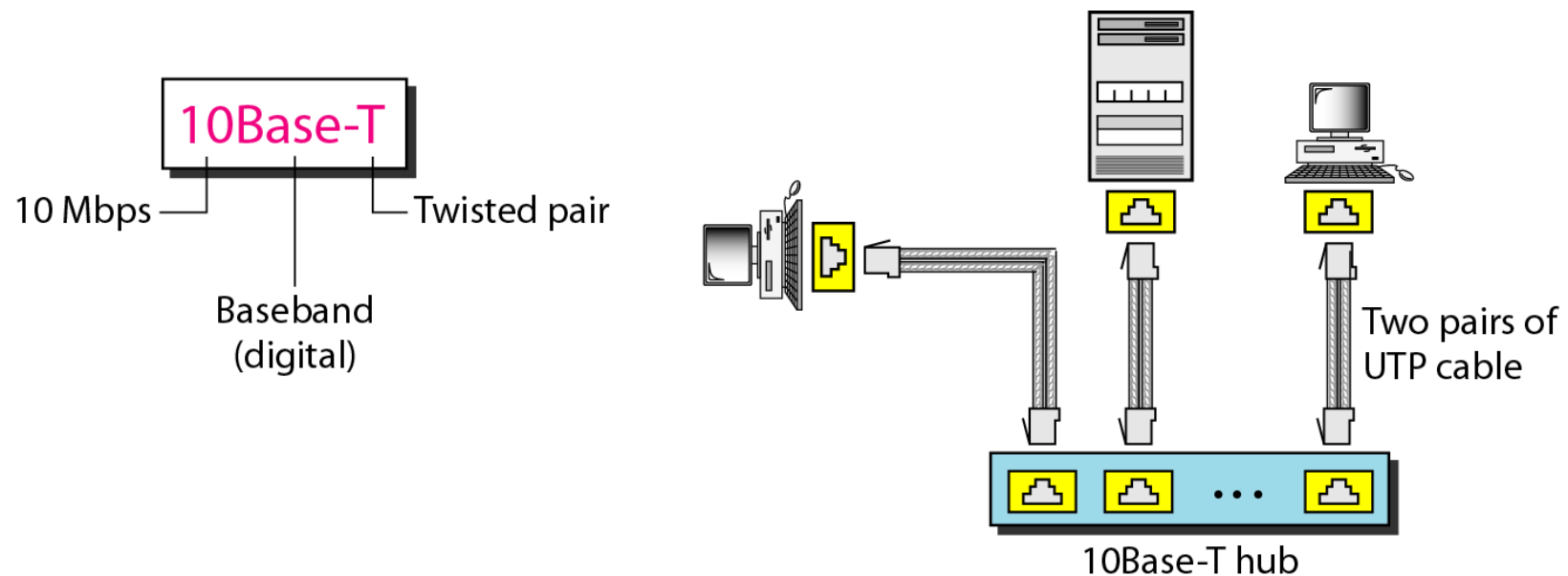
10Base2: Thin Ethernet

- Uses a bus topology, but the cable is much thinner and more flexible
- The transceiver is normally part of the network interface card (NIC), which is installed inside the station
- Thin coaxial cable is less expensive than thick coaxial
- The length of each segment cannot exceed **185 m** (close to 200 m) due to the high level of attenuation in thin coaxial cable



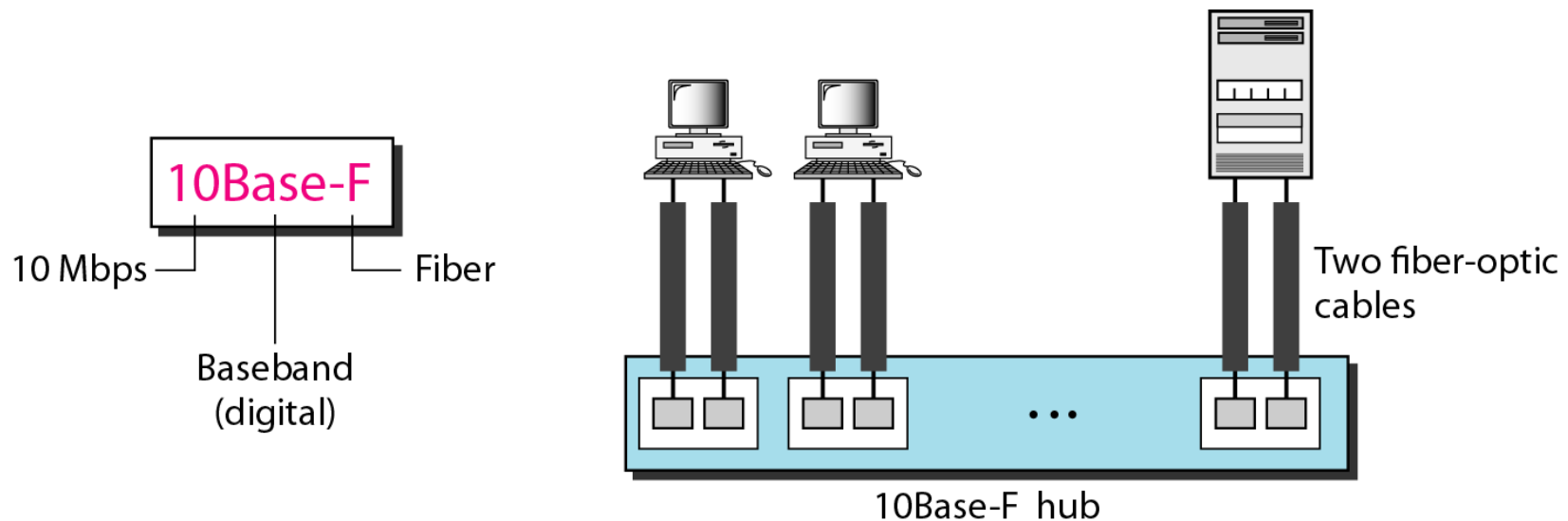
10Base-T: Twisted-Pair Ethernet

- A physical **star topology**
- The stations are connected to a hub via two pairs of twisted cable
- Any collision here happens in the hub
- The maximum length of the twisted cable here is defined as **100 m**



10Base-F: Fiber Ethernet

- Uses a **star topology** to connect stations to a hub.
- The stations are connected to the **hub** using two fiber-optic cables
- Maximum length **2000m**



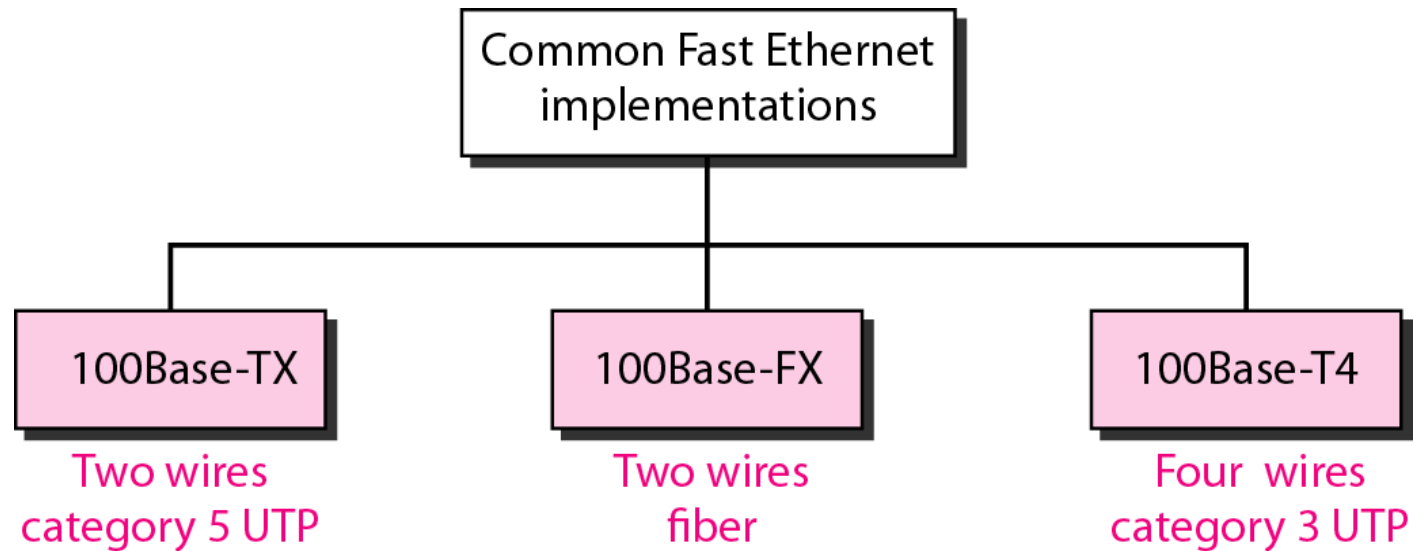


Fast Ethernet

- IEEE 802.3u standard
- Upgrade the data rate to 100 Mbps.
- Make it compatible with Standard Ethernet.
- Point-to-point or star topology
- Keep the same 48-bit address.
- Keep the same frame format.
- Keep the same minimum and maximum frame lengths



Fast Ethernet



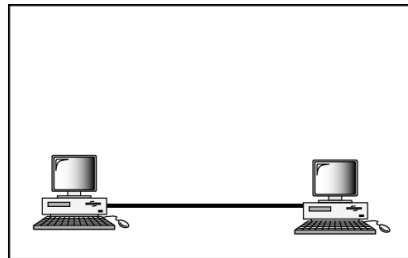
<i>Characteristics</i>	<i>100Base-TX</i>	<i>100Base-FX</i>	<i>100Base-T4</i>
Media	Cat 5 UTP or STP	Fiber	Cat 4 UTP
Number of wires	2	2	4
Maximum length	100 m	100 m	100 m
Block encoding	4B/5B	4B/5B	
Line encoding	MLT-3	NRZ-I	8B/6T



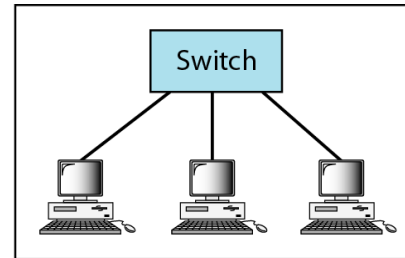
Gigabit Ethernet

- IEEE Standard 802.3z
- Upgrade the data rate to 1 Gbps.
- Make it compatible with Standard or Fast Ethernet.
- Use the same 48-bit address.
- Use the same frame format.
- Keep the same minimum and maximum frame lengths.
- To support autonegotiation as defined in Fast Ethernet
- In the full-duplex mode of Gigabit Ethernet, there is no collision;
- the maximum length of the cable is determined by the signal attenuation in the cable.

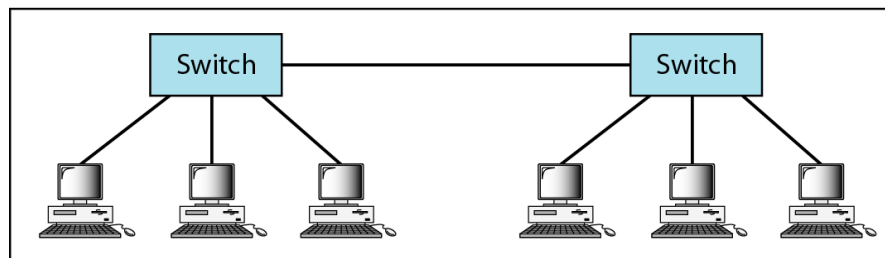
Gigabit Ethernet topologies



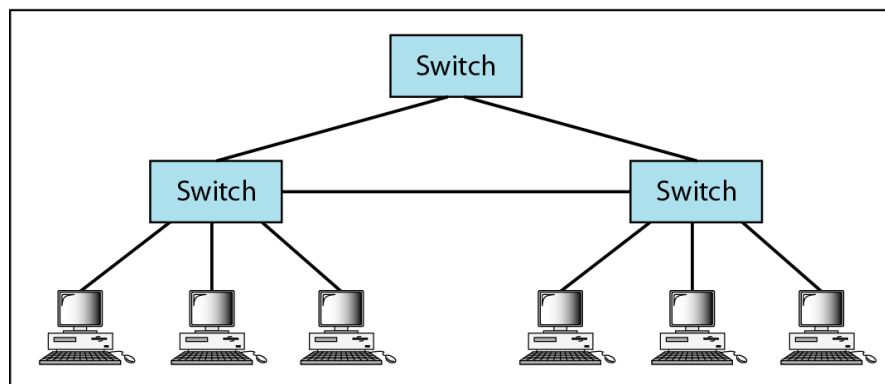
a. Point-to-point



b. Star

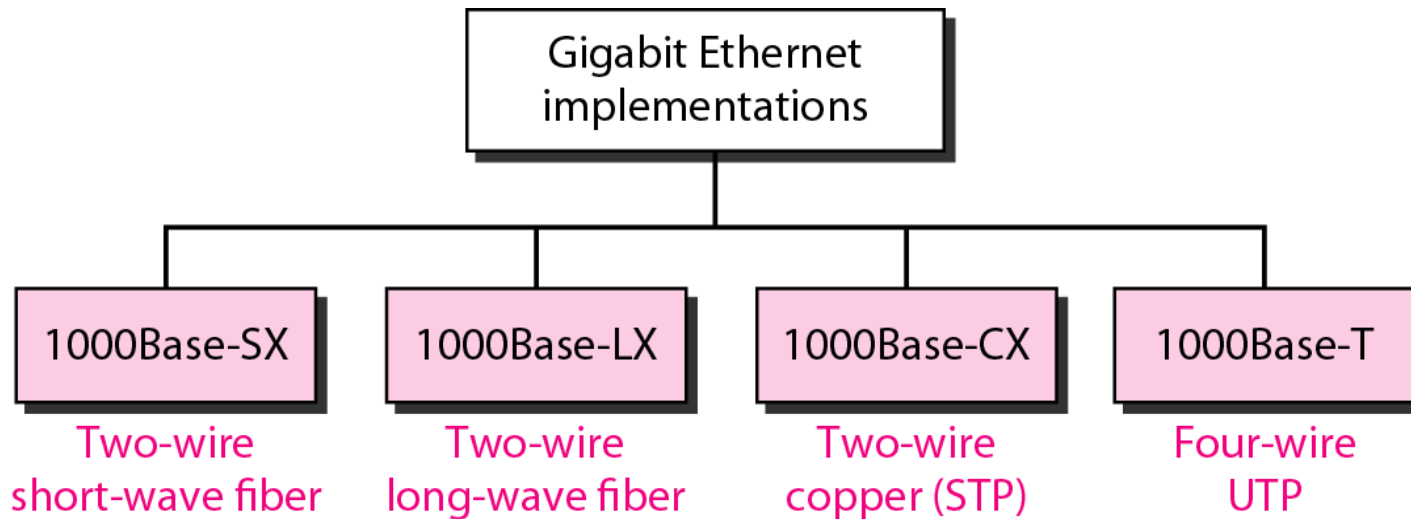


c. Two stars



d. Hierarchy of stars

Implementations



Characteristics	1000Base-SX	1000Base-LX	1000Base-CX	1000Base-T
Media	Fiber short-wave	Fiber long-wave	STP	Cat 5 UTP
Number of wires	2	2	2	4
Maximum length	550 m	5000 m	25 m	100 m
Block encoding	8B/10B	8B/10B	8B/10B	
Line encoding	NRZ	NRZ	NRZ	4D-PAM5



Ten-Gigabit Ethernet

- IEEE Standard 802.3ae
- Upgrade the data rate to 10 Gbps.
- Make it compatible with Standard, Fast, and Gigabit Ethernet.
- Use the same 48-bit address.
- Use the same frame format.
- Keep the same minimum and maximum frame lengths.
- Allow the interconnection of existing LANs into a metropolitan area network (MAN) or a wide area network (WAN).
- Make Ethernet compatible with technologies such as Frame Relay and ATM

<i>Characteristics</i>	<i>10GBase-S</i>	<i>10GBase-L</i>	<i>10GBase-E</i>
Media	Short-wave 850-nm multimode	Long-wave 1310-nm single mode	Extended 1550-nm single mode
Maximum length	300 m	10 km	40 km



Next : Connecting devices and VLAN